

A Lightweight Deep Learning Model for Real-time Detection and Recognition of Traffic Signs Images Based on YOLOv5

Hui He¹, Qihong Chen¹, Guoping Xie², Boxiong Yang^{*3}, Shelei Li³, Bo Zhou³, Yuye Gu³

¹Advanced Institute of Natural Science, Beijing Normal University, Zhuhai, China

²China Information Center, Beijing, China

³School of Information & Intelligence Engineering, University of Sanya, Sanya, China

*Correspondence: boxiongyang@sanyau.edu.cn

Abstract—The rapid and accurate identification of various road traffic signs is an important research topic in automotive vision systems. Specifically, the correct identification of road signs is an urgent problem requiring effective solutions to facilitate automatic driving. This paper proposes a new approach, PP-LCNet-P2-CT, for the detection and recognition of urban road signs in an automotive vision system using an improved YOLOv5 deep learning model. The main improvement of the PP-LCNet-P2-CT model includes the following: (1) Replacing the YOLOv5 backbone network with the lightweight network PP-LCNet to improve the real-time performance of the detection network; (2) Adding a small target detection head to the detection head to meet the needs of target detection with different scales and mitigate the adverse effects caused by drastic target scale changes; and (3) Integrating the CBAM convolutional block attention model that focuses on target features and the transformer coding block that can capture different local information to ensure the accuracy of lightweight model target detection. The model was tested with the Tsinghua traffic sign dataset, TT100k. The results show that the mAP index of the PP-LCNet-P2-CT model is increased by 29.84% and the FPS is increased by 24.05%, while the number of model parameters is decreased by 32.78% and the GFLOPs decreased by 34.41% compared with the classic YOLOv5 algorithm. The PP-LCNet-P2-CT model allows complex deep learning to be used successfully for unmanned ground vehicles (UGVs) with ordinary computing speeds and high real-time requirements.

Keywords—real-time detection and recognition; road traffic signs; attention mechanism; improved deep learning model; YOLOv5

I. INTRODUCTION

Currently, a global need exists to improve the vehicle safety [1]. More than 90% of global vehicles crashes are due to driver error [2]. Intelligent vehicles have the capability to improve vehicle safety by assisting or removing the driver completely. In the near future, the current levels of driving assistance will be replaced by fully autonomous vehicles, which will drastically improve safety and reduce human operation errors.

In recent decades, vision-based vehicle detection algorithms have been used in driver assistance systems. The identification of road traffic signs is one of the key technologies for intelligent vehicle driving in complex traffic situations. Therefore, effectively and correctly identifying road traffic signs is an urgent problem for unmanned ground vehicles (UGVs).

Traditional traffic sign detection methods mainly rely on the color and shape of traffic signs and depend on the inherent red,

blue and yellow colors and specific shape characteristics of traffic signs [3]. With the development of artificial intelligence technology, some researchers have proposed traffic sign detection algorithms based on convolutional neural networks (CNNs), which are divided into two-stage target detection algorithms based on candidate regions and single-stage target detection algorithms based on regression [4].

Although the two-stage method based on candidate regions has good detection and positioning effects, the two-stage method based on candidate regions cannot meet the real-time requirements of practical applications. Thus, the single-stage method based on regression is usually used.

Huang Z. et al. [5] present a milestone leap in the application of convolutional neural networks to target detection; it describes the use of a CNN network to extract the features of candidate regions from top to bottom for target detection and segmentation.

Girshick R. D. et al. [6] proposes a pyramid network. The author adds a feature pyramid pooling layer between the last convolution layer and the full connection layer, which allows pictures of any size to be input. The research in [7] describes the use of a deep convolutional neural network for efficient target detection based on R-CNN, which uses VGG-16 network. This network is 9 times faster than R-CNN in training and 213 times faster in testing, and achieves higher mAP on PASCAL VOC 2012. Further, Authors in [8] introduce the region proposal network (RPN) to replace the selective search network, which reduces the detection time by approximately 10 times under the condition of ensuring the accuracy of object detection and realizes real-time object detection.

In addition, the efforts in [9] propose the feature pyramid network (FPN) model. The FPN not only predicts each feature map but also performs top-down feature fusion, which fuses high-level semantic information with low-level high-resolution information, improves the expression ability of features, and improves object detection.

Wu, Y et al. [10] proposes the differencing R/B algorithm and Retinex algorithm to segment the traffic sign candidate area and the sky area, respectively. In terms of color segmentation, a threshold is used to remove background interference, and the dark light is removed in the nonsky area to enhance image preprocessing.

This paper was supported by the Zhuhai Foundation and Application Foundation Research Project (No. ZH22017003200027PWC), Hainan Province Science and Technology Special Fund (No. ZDYF2022GXJS005).

Moreover, an image restoration CNN (IRCNN) algorithm is proposed in [11] such that IRCNN can be used to first remove haze, and the multichannel convolutional neural network can then be used to detect and recognize the image after haze removal.

Differently, Zhang H. et al. [12] proposed a modular architecture separation attention module, which weights different channels by adding the separation attention module. Channel attention is applied to different network branches to realize the cross group of the attention mechanism for feature maps in the network.

We improve YOLOv5 design for the purpose of real-time traffic sign detection and identification. The improvement includes the backbone design and head with an attention method. The YOLOv5 backbone network is replaced by the lightweight network PP-LCNet. A small target detection head is added to the detection head. Meanwhile, CBAM convolutional block attention model and Transformer coding block are integrated. Therefore, the problems such as difficult identification of small targets and large weight of the model are solved, and the lightweight and high precision of the traffic sign detection model are completely implemented. The final experimental results are demonstrated in the paper the improvement of deep learning efficiency based on the PP-LCNet-P2-CT model we propose.

II. MATERIALS AND METHODS

YOLOv5 is a family of object detection architectures and models pretrained on the COCO dataset [13]. The YOLOv5 network is divided into a backbone network, neck network and prediction head. A backbone network is a convolutional neural network that aggregates and forms image features with different granularities. The neck is located between the backbone network and the prediction head. It is a series of layers used for mixing and combining image features to better utilize the characteristics

of the backbone network. The prediction head uses features received from the neck and predicts frames and categories. The YOLOv5 network uses CSPDarknet-53 [14] +SPP [15] as the network backbone, and the PANet structure is the neck and YOLO prediction head.

Although YOLOv5 is excellent in deep learning target recognition, especially in small target recognition, its network structure is too complex and computation requirement is too large, making it unsuitable for real-time processing in UGVs systems with limited computing resources [16]. Therefore, we selected the lightweight model PP-LCNet as the backbone network [17]. To fuse more features without increasing the computational cost, PANet was adopted as the main neck architecture. In addition, a prediction head for detecting small targets was added to improve the detection performance of small targets [18]. At the same time, the model also integrates the Convolutional Block Attention Model (CBAM) which helps the network find the region of interest in dense scenes, and applies Transformer coding block to the prediction header to tap the potential of feature representation [19].

The main advantage of the improved YOLOv5 is that it can be applied to UGVs systems, whereas most of the networks are unsuitable for this application because of the heavy model weight and complex network structure. In addition to considering accuracy and real-time needs, the size of the model parameters also needs to be balanced.

A. Network model design

The model architecture is shown in Fig.1. The backbone network is a structure similar to that of PP-LCNet-x-1. The neck adopts the PANet structure as the main architecture, and the prediction head adds a detection head of small targets and integrates the CBAM convolutional block attention model and transformer coding block.

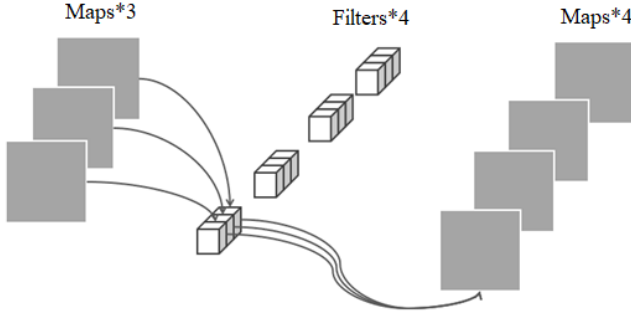


Fig. 4. PW Convolution

The SE module [22] can balance network channels to help us obtain better results, but it also increases the reasoning time of the model. Therefore, we chose to add the SE module to the module near the end of the network after balancing the accuracy and speed of the model. That is, we only used the SE module on the deep separable convolution block with 1024 output channels and 5 convolution cores in the last two layers. At the end of the backbone network, SPP-Fast was used to produce a fixed dimension feature map and pass it into the neck feature extraction network of the model.

C. Neck module

The Neck module follows the original PANet structure of YOLOv5 [23] and uses a feature fusion strategy (see Fig.5(b)). As the main architecture, the neck module performs multiscale fusion on the features extracted from the backbone network, combines the shallow detail information with the deep semantic information, and enhances the feature extraction ability.

The Neck module has an additional bottom-up path compared with the top-down structure of FPN (see Fig.5(a)). Moreover, it adds a multilayer shortcut between the top-down module and the bottom-up module so that it can fuse more features without increasing the computational cost. In addition, a prediction head for detecting micro-targets was added to improve the ability of the model to detect targets of different scales. At the same time, the model also integrates the convolutional attention module CBAM to find the attention area in the image with large coverage. In addition, this model also carries out some applications of transformer encoder blocks in prediction headers. In combination with YOLO's original three detection layers, the newly added tiny target detection layer P2 can help alleviate the adverse effects caused by drastic target scale changes. The new prediction head is generated from the low-level high-resolution feature image, which can help detect small targets. For example, when the input image size is 640×640 , the newly added detection layer size is 160×160 , which can be used to detect targets with a size of more than 4×4 .

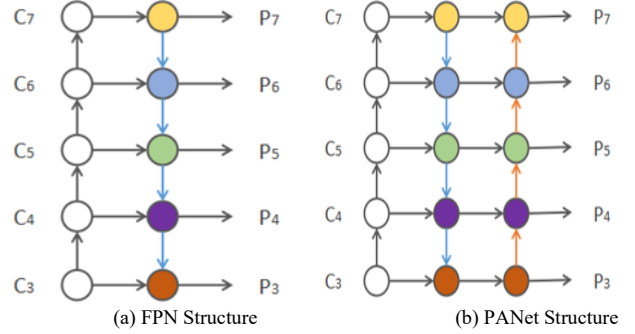


Fig. 5. Structure of the Neck module

D. CBAM

The convolutional block attention module (CBAM) is a lightweight module that can be trained in an end-to-end manner and integrated into the CNN architecture [24]. The calculation process (see Fig.6) is as follows:

First, a feature image is input, and CBAM will then carry out attention mapping along the two dimensions of channel and space and multiply it with the input feature image to refine adaptive features. Finally, the optimized features are obtained. The CBAM convolutional attention module applied in the model can avoid the interference of geographical elements or other backgrounds so that the model focuses on the target object. Therefore, CBAM is used to strengthen the features of key areas and reduce invalid interference features before each detection layer.

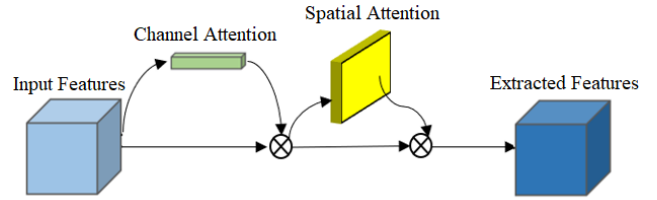


Fig. 6. Operation process of CBAM

Each transformer encoder block (see Fig. 7) contains two sub layers, and each sublayer is connected by a residual. The first layer is the multi-head attention layer, and the second layer is the fully connected layer. The transformer encoder block enhances the ability to capture different local information and can also use a self-attention mechanism to accurately locate targets in high-density scenes. In addition, the transformer encoder block can capture global information and rich context information and performs better for high-density occlusion objects. The model proposed in this paper only adds the transformer encoder block to the end of the network because the resolution of the feature map at the end of the network is low, and the use of the transformer encoder block structure can maximize the reduction of computing resources.

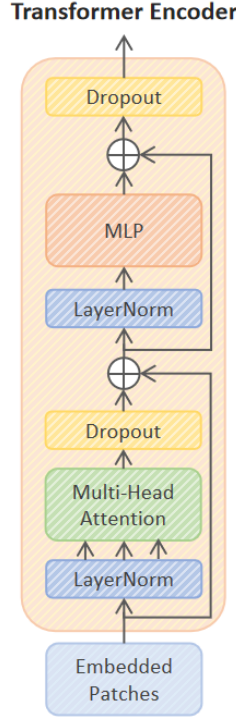


Fig. 7. Structure of Transformer Encoder Block

III. THE EXPERIMENT

A. Dataset

The public dataset TT100K (Tsinghua Tencent 100k) selected in this experiment contains road traffic sign information based on real scenes in the urban and suburban areas of five different cities in China. This dataset is compiled and released by the Joint Laboratory of Tsinghua University and Tencent company. The TT100K dataset includes 151 types of traffic signs, including three major categories: instruction, prohibition and warning, which are divided into 128 subcategories (see Fig. 8). TT100K contains a total of 9176 images of 30000 traffic sign instances, and the pixels of each image are 2048×2048 . Specifically, 6105 of these images were used for training, and 3071 were used for testing. TT100K contains many pictures of traffic signs in complex scenes, and the sides on which traffic sign instances appear also markedly differ. These images were collected from natural scenes under different conditions with a large base and rich semantics.

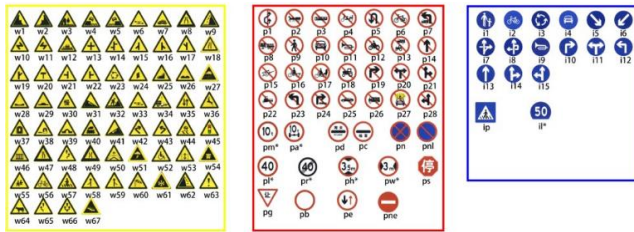


Fig. 8. Examples of road traffic signs in the TT100K dataset

According to statistics (see Fig.9), TT100K contains 106 categories for which the number of instances is less than 50. In this study, 9170 pictures in 45 categories with more than 50 instances were selected as the basic dataset to prevent serious data distribution imbalances and overfitting. The basic dataset was then divided into a training dataset and a test dataset at a ratio of 9:1.

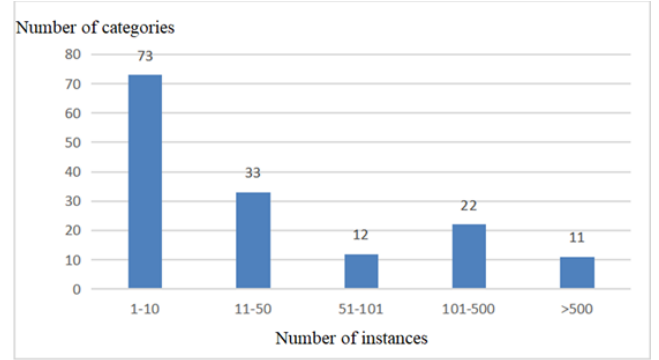


Fig. 9. Instances-category statistics of TT100K

B. Data augmentation processing

The model proposed in this paper was needed to convert the dataset from TT100K format to VOC format. After successful conversion, the VOC format label must be converted into the required dataset of YOLO format.

In addition, the dataset was also enhanced during the experiment. The data enhancement strategies included HSV transformation, image deformation, blending and stitching, as shown in Fig.10.



Fig. 10. Image augmentation example

C. The Experimental design

The results of different algorithms to train the model under the same training parameters were compared to ensure the accuracy of the experimental results. Specific training parameters and parameter values are shown in TABLE I.

TABLE I. CONFIGURATION TABLE OF EXPERIMENTAL PARAMETERS

Training parameters	Parameters Value
Weight Decay	0.0005
Momentum	0.937
Batch size	4
lr0 (Initial learning rate)	0.0032
lrf (Cycle learning rate)	0.1
warmup_epochs	3.0
warmup_bias_lr	0.1
warmup_momentum	0.8

D. The Experimental design

The four parameters of mAP, FPS, parameters, and GFLOPs were taken as the reference indices of the model experiment results in this study.

Precision is the accuracy rate, and it was calculated as follows: the actual number of positive samples in the predicted samples/the number of all positive samples, as in (1):

$$Precision = \frac{TP}{TP+FP} \quad (1)$$

Recall is the recall rate, and it was calculated by dividing the number of actual positive samples by the number of predicted samples in the predicted samples, as in (2):

$$Recall = \frac{TP}{TP+FN} = TP/P \quad (2)$$

IV. RESULTS AND DISCUSSION

A. Ablation experiment

A group of ablation experiments were devised in this study to verify the effect of the algorithm on adding small object detection heads and evaluate improvements in the accuracy of the algorithm in response to the addition of transformer coding blocks and CBAM modules. The ablation experiments were designed as follows:

- ① PP-LC-YOLOv5x Model after replacing the backbone network;
- ② PP-LC-YOLOv5x-P2 Model of micro-object detection head is only added;
- ③ PP-LC-YOLOv5x-P2 Model with only the CBAM module embedded;
- ④ PP-LC-YOLOv5x-P2 Model with only transformer coding block added;
- ⑤ The PP-LC-YOLOv5x-P2 Model with both the CBAM module and transformer coding block is added.

The above five models were trained on the TT100K dataset under the same experimental environment, including the same software and hardware, data augmentation and training strategy. The performance results of the ablation experiment are shown in Table II.

TABLE II. PERFORMANCE INDEX OF THE ABLATION EXPERIMENT

Model	Parameters	GFLOPs	mAP@.5	FPS
PP-LC-YOLOv5x	47708450	95.5	0.328	33.44
PP-LC-YOLOv5x-P2	49374200	132.5	0.348	23.98
PP-LC-P2-CBAM	49505890	132.8	0.359	22.08
PP-LC-P2-Trans	57991800	134.2	0.389	23.36
PP-LCNet-P2-CT	58123490	134.4	0.397	21.46

As shown in Table II, the accuracy of Model ② was improved by 5.7% compared with Model ①; although the parameter quantity was increased by 3.4%, the calculation quantity was increased by 27.9%, and the detection speed was decreased by 28.29%.

The accuracies of Model ③ and Model ④ improved by 3.1% and 10.5%, respectively, compared with model ②. Although the parameters of Model ③ and Model ④ increased by 0.3% and 14.9%, respectively, the calculation amounts increased by 0.2% and 1.3%, respectively, and the detection speeds decreased by 7.9% and 2.6%, respectively.

Compared with Model ②, the precision of Model ⑤, which added the small object detection head, CBAM module and transformer coding block, was improved by 12.3%; although the parameter amount was increased by 15.1%, the calculation amount was increased by 1.4%, and the detection speed was decreased by 10.5%.

In summary, the precision of the improved model (PP-LCNet-P2-CT) in this paper is better than that of the model with only the CBAM module and trans coding block added, but this improvement came at the expense of the number of parameters, the cost of computation and the detection speed.

B. Comparison of model scale

The scale was compared between the improved PP-LCNet-P2-CT proposed in this paper and YOLOv5x in the three aspects (memory, parameters, and GFLOPs). From Table III, we can conclude that the memory required by the improved PP-LCNet-P2-CT model in this paper is 112.2 MB, which is 32.29% less than that of the YOLOv5x model. The parameter quantity is 58,123,490, which is 32.78% less than that of the YOLOv5x model. The GFLOPs are 134.4, which is 34.41% less than that of the YOLOv5x model. The above data show that the model proposed in this paper reduces the complexity of the model and is more conducive to the practical application of road traffic sign detection and recognition for UGVs.

TABLE III. COMPARISON RESULTS OF MODEL SCALE

Model	Memory (MB)	Parameters	GFLOPs
YOLOv5x	165.7	86469490	204.9
PP-LCNet-P2-CT	112.2	58123490	134.4

C. Comparison of model performance

The performances of the improved PP-LCNet-P2-CT model proposed in this paper and the YOLOv5x model were compared. The experimental results of the two models on the

TT100K dataset are shown in Table IV. The mAP@.5 of the model proposed herein is 0.573, an increase of 29.84% compared with the mAP@.5 of .402 for YOLOv5x. In addition, the FPS of the model in this paper is 21.83, which is increased by approximately 24.05% compared with FPS of 16.58 for the YOLOv5x model.

TABLE IV. COMPARISON OF MODEL PERFORMANCE

Model	mAP@.5	mAP@.5:.95	FPS
YOLOv5x	0.402	0.289	16.58
PP-LCNet-P2-CT	0.573	0.425	21.83

V. CONCLUSIONS

In this paper, we proposed an improved PP-LCNet-P2-CT for the YOLOv5 deep learning model. PP-LCNet-P2-CT replaces the YOLOv5 backbone network with the lightweight network PP-LCNet. The part of Neck follows the structure of PANet and adds a small target detection head to the detection head part. Meanwhile, it integrates the CBAM convolution block attention model and the transformer coding block.

Although the PP-LCNet-P2-CT algorithm featured highly accurate performance in real-time for the TT100k dataset, the actual application of the algorithm is still associated with many shortcomings. For example, the recognition effect of the model is not good in real traffic roads when the model is blocked by plants, buildings or other objects in bad weather or when the traffic signs themselves are seriously damaged.

The algorithm may be further improved as follows:

- (1) The detection and recognition of road traffic signs in extreme and harsh natural environments can be improved;
- (2) The reduced accuracy of detection and recognition due to the fading or defacement of road traffic signs as a result of long-term exposure to the outdoor environment can be improved;
- (3) Inaccuracies in detection and recognition caused by motion blur during vehicle driving can be improved.

ACKNOWLEDGMENT

The paper was supported by the Zhuhai Foundation and Application Foundation Research Project (No. ZH22017003200027PWC), Hainan Province Science and Technology Special Fund (No. ZDYF2022GXJS005). The authors gratefully acknowledge the reference authors, the Graphics and Image Public Laboratory of Zhuhai, the Intelligent Engineering and Educational Application Research Centre, and the Advanced Research Institute of Natural Sciences of Beijing Normal University, Supercomputing Center of University of Sanya, China.

REFERENCES

[1] Song, Y., Fan, R., Huang, S. et al. A three-stage real-time detector for traffic signs in large panoramas. *Comp. Visual Media*, 2019.5, pp. 403-416.
[2] G. Sabaliauskaite, L. S. Liew, F. Zhou and J. Cui. Designing Safe and Secure Mixed Traffic Systems [C]. 2019 IEEE 19th International Symposium on High Assurance Systems Engineering (HASE), 2019, pp. 222-227.

[3] Ma Yongjie, Li Xueyan, Song Xiaofeng. Traffic sign recognition based on improved deep convolution neural network [J]. *Laser & Optoelectronics Progress*, 2018,55(12), pp. 250-257.
[4] Cai Zixing, Gu Mingqin. Traffic sign recognition algorithm based on shape signature and dual-tree complex wavelet transform [J]. *Journal of Central South University*, 2013, 20(2), pp. 433-439.
[5] Huang Zhiqiang, Li Jun. Research on traffic sign recognition based on spatial channel attention mechanism and multi-scale fusion[J]. *Journal of Nanjing University of Posts and Telecommunications (Natural Science Edition)*, 2022,42(02), pp.93-102.
[6] Girshick, R., Donahue, J., Darrell, T., et al. Rich feature hierarchies for accurate object detection and semantic segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2014, pp. 580-587.
[7] He K , Zhang X , Ren S , et al. Spatial Pyramid Pooling in Deep Convolutional Networks for Visual Recognition[J]. *IEEE Transactions on Pattern Analysis & Machine Intelligence*, 2014, 37(9), pp. 1904-16.
[8] Ren S , He K , Girshick R , et al. Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks[J]. *IEEE Transactions on Pattern Analysis & Machine Intelligence*, 2017, 39(6), pp. 1137-1149.
[9] Lin T Y , Dollar P, Girshick R , et al. Feature Pyramid Networks for Object Detection[C]// 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). IEEE Computer Society, 2017, pp. 936-944.
[10] Wu, Y.; Tang, S.; Zhang, S.; Ogai, H. An Enhanced Feature Pyramid Object Detection Network for Autonomous Driving. *Appl. Sci.* 2019, 9, pp. 4363.
[11] Chen Xiuxin, Ye Yang, Yu Chongchong, et al. Identification of Traffic Signs in Haze Weather Based on Deep Learning[J]. *Journal of Chongqing Jiaotong University. Natural Science*, 2020,39 (12), pp.1-5,12.
[12] Zhang H, Wu C, Zhang Z, et al. Resnet: Split-attention networks [J]. 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2022, pp. 2735-2745.
[13] Liu W J , Liang X J , Qu H C. Learning performance of convolutional neural networks with different pooling models. *Journal of Image and Graphic*, 2016,21(9) , pp. 1178-1190.
[14] Wang C Y, Liao H Y M, Wu Y H, et al. CSPNet: A new backbone that can enhance learning capability of CNN, In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*. 2020, pp. 390-391.
[15] Huang Z, Wang J, Fu X, et al. DC-SPP-YOLO: Dense connection and spatial pyramid pooling based YOLO for object detection[J]. *Information Sciences*, 2020, 522, pp. 241-258.
[16] Liu S, Qi L, Qin H, et al. Path aggregation network for instance segmentation[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2018, pp. 8759-8768.
[17] Cui C, Gao T, Wei S, et al. PP-LCNet: A Lightweight CPU Convolutional Neural Network[J]. *arXiv preprint*, 2021.
[18] Woo S, Park J, Lee J Y, et al. CBAM: Convolutional block attention module[C]//Proceedings of the European conference on computer vision (ECCV). 2018, pp. 3-19.
[19] Dosovitskiy A, Beyer L, Kolesnikov A, et al. An image is worth 16x16 words: Transformers for image recognition at scale[J]. *arXiv preprint* , 2020.
[20] Ashish Vaswani, Noam Shazeer, Niki Parmar, et al. Attention is all you need [J]. 31st Conference on Neural Information Processing Systems (NIPS 2017), 2017, pp. 1-15.
[21] Wang K , Liew J H , Zou Y , et al. PANet: Few-Shot Image Semantic Segmentation with Prototype Alignment [C]. 2019 IEEE/CVF International Conference on Computer Vision (ICCV), 2019, pp. 9196-9205.
[22] Chollet F. Xception: Deep learning with depthwise separable convolutions[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2017, pp. 1251-1258.
[23] J. Hu, L. Shen, S. Albanie, G. Sun and E. Wu. Squeeze-and-Excitation Networks[J]. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2020, 42(8), pp. 2011-2023.
[24] Zhang H, Cisse M, Dauphin Y N, et al. mixup: Beyond empirical risk minimization[J]. 2017.
[25] Yun S, Han D, Oh S J, et al. Cutmix: Regularization strategy to train strong classifiers with localizable features, In *Proceedings of the IEEE/CVF international conference on computer vision*, 2019.